groups and hashed B-Tree lookups in large directories. Uninitialised block groups are instrumental in enabling faster fsck and the details may be found at *http://kerneltrap.org/Linux/Improving_fsck_Speeds_in_Ext4*.

You will need to run *fsck* before you can use the filesystem, as the data structures need to be rebuilt before you can mount them as an ext4 filesystem. However, backing up your data is always a good practice! Note that you won't be able to mount */dev/foo* as an ext3 filesystem again!

Mount an existing ext3 FS

If your kernel has support for ext4, you can *mount* your existing ext3 filesystems as ext4, using the usual mount syntax. However, you will not be able to make use of ext4 features, which require data structure modifications such as extents, fast fsck, etc. You would be able to take advantage of delayed allocation or the multiple block allocator though, which do not change the data structures of the file system. The allowed mount options are described in the documentation for ext4, mentioned in the *Resources* section.

Distribution-specific ways to use ext4 filesystems are listed in the ext4 *how-to* mentioned under *Resources*.

Epilogue

I hope this article has served its purpose of being a primer on ext4, while highlighting its advantages over its predecessors. The new features list is pretty varied, and so there is probably at least one reason to try out ext4. Unless you try, you are sure to miss out on the advancements in the world of the Linux kernel!

The source code of ext4 resides in the *fs/ext4/* sub-directory of the kernel source tree. You know where to go after reading this article: *www.kernel.org*. END

Resources

- ext4 documentation: *http://lxr.linux.no/linux+v2.6.32/Documentation/filesystems/ext4.txt*
- ext4 How-to: *http://ext4.wiki.kernel.org/index.php/Ext4_Howto*
- Kernel newbies page on ext4: *http://kernelnewbies.org/Ext4*
- Anatomy of ext4: *www.ibm.com/developerworks/linux/library/l-anatomy-ext4*
- Migrating to ext4: *www.ibm.com/developerworks/linux/library/l-ext4 (Pretty outdated, but relevant.)*
- Details of multi-block allocation and other changes in inode/allocator: *http://ols.fedoraproject.org/OLS/Reprints-2008/kumar-reprint.pdf*
- Fast fsck: *http://kerneltrap.org/Linux/Improving_fsck_Speeds_in_Ext4*
- Wikipedia entry on ext4: *http://en.wikipedia.org/wiki/Ext4*

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Continued from page 42

Upstart is a replacement for SysVinit, and is an asynchronous event-based daemon that initialises services simultaneously. It was written by Canonical employee Scott James Remnant, and made its debut in Ubuntu 6.10 Edgy Eft. In itself, Upstart has perfect backward compatibility with SysVinit and can use its scripts natively, making it a great drop-in replacement.

However, to fully realise the power of Upstart, one has to write Upstart Event Definitions. UEDs are simple text files (actually just basic .conf files) that are non-executable. They just list the executable file, specify any scripts to run, if necessary, and leave the rest of the housekeeping work to Upstart. As yet, only Ubuntu 9.10 Karmic Koala uses UEDs, that too, only in the core bootstrapping sequence. UEDs made a debut in Karmic only in Alpha 6. The result, as we all know, is a blazingly fast boot.

All we need now is to wait for Upstart Event Definitions to replace the *init* scripts in Fedora (as well as other distros), and we'll be opening that presentation on our desktop even before we hit the power button... almost literally! Gosh, does that mean we'll have less time to look at that pretty boot splash screen ;-)

Next up...

Well, the infrastructure is now in place. All it needs is a bit of tweaking to get it to where the system boots in less than 15 seconds, even on stupid hardware.

There's this unresolved bug (yes, it's a big one), which causes Plymouth to be the first job to get killed when the system is shutting down, creating a black screen (but it's still flicker-free) and dampening the shutdown experience. This thing will be corrected once UEDs make an appearance in Fedora. That's the only way to fix it without using what we call ugly hacks. END

References

- Dracut project home: *http://sourceforge.net/apps/trac/dracut/wiki*
- Fedora wiki on Dracut: *http://fedoraproject.org/wiki/Dracut*
- Fedora wiki on Initrdrewrite: *https://fedoraproject.org/wiki/Initrdrewrite*
- Wikipedia on Mode-setting: *http://en.wikipedia.org/wiki/Mode-setting*
- From Power Up To Bash Prompt: *http://tldp.org/HOWTO/From-PowerUp-To-Bash-Prompt-HOWTO.html*

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